

Natalie McDonald

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EDUCATION:

Virginia Tech, Blacksburg VA

BS in Mechanical Engineering – Spring 2027

Majoring in Robotics and Mechatronics

Cumulative GPA: 3.13

SKILLS:

MATLAB, Simulink, SolidWorks, Autodesk Fusion,
Python, 3D Printing, Microsoft Office Suite, Mathematica,
C/C++, Siemens NX

WORK EXPERIENCE:

Virginia Tech American Society of Mechanical Engineers (ASME) Outreach Coordinator

Starting June 2026

Incoming M.C. Dean Mechanical Engineering Intern, Tysons VA

June 2026 to Aug 2026

3DPAC Finance Lead

Starting June 2026

- Manage team expenses by establishing a budget and securing funding through sponsorships and external partnerships
- Coordinate outreach events to showcase the team's work and oversee the team's Instagram presence

Army DEVCOM Chemical Biological Center Internship, Edgewood MD

June 2025 to Sept 2025

- Support the Sustainment and Acquisition Branch to design and execute an experimental test plan to evaluate additive manufacturing materials under repeated decontamination cycles for potential military applications.
- Collaborate with engineers to document findings and support decisions for future procurement efforts.

Virginia Tech Undergraduate Teaching Assistant, Blacksburg VA

Aug 2025 to Present

- Provide academic support for ME2004, Engineering Analysis with Numerical Methods, by assisting students in applying MATLAB and Python based computational methods to solve real-world problems.
- Host weekly office hours to mentor students, clarify concepts, and foster collaborative problem solving.

Virginia Tech Software Assistance and Triage Student Technician, Blacksburg VA

Jan 2025 to Present

- Assist engineering students with installing and troubleshooting software products while providing immediate assistance to student requests through staffing a virtual help desk.
- Conduct hardware and software diagnostics to identify and resolve computer problems.

GrayUR Interdisciplinary Undergraduate Research Group, Blacksburg VA

Sept 2024 to May 2025

- Collaborated with the Army at Ft. Belvoir and a team of undergraduate student to design and fabricate a low-cost airframe capable of supporting a multispectral camera payload for military applications.
- Contributed to restructuring the project to align with new goals and selecting efficient materials for the fuselage.

National Security Agency Engineering Technician Intern, Ft. Meade MD

Sept 2022 to Aug 2023

- Collaborated on operational tasks and lifecycle repairs, incorporating specialty equipment such as SIGINT systems
- Passed a Specialty Background Investigation, including a full-scope polygraph, and received security clearance.

RELATED ACTIVITIES AND PROJECTS:

MachWorks Design Team at Virginia Tech

June 2024 to Jan 2026

- Build expertise in MATLAB, Simulink, and Python by working with the propulsion sub-team to develop a dynamic model of the engine, characterizing the relationship between throttle control position and thrust output.
- Developed a program with Python to record the movement of a rotary encoder and model the position of the servo.

3D Printed Aircraft Competition (3DPAC) at Virginia Tech

Feb 2025 to Present

- Advance manufacturing skills as a member of the print sub-team to translate design criteria into physical parts.
- Optimized CAD models for additive manufacturing by refining geometries, selecting appropriate print parameters, and troubleshooting print failures to ensure structural integrity and weight efficiency of aircraft components.
- 3DPAC won first place for Innovation at the annual competition at University of Texas Arlington last year.